CS-4063: Natural Language Processing

Assignment 2 — Neural NLP Pipeline

A Continuation of the BBC Urdu NLP Pipeline

Assingment Report

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DS-A

Part 1: Word Embeddings:

1. TF-IDF and PMI Weighted Representations:

1.1 TF-IDF Weighting:

Top-10 most discriminative words per topic category using TF-IDF scores:

TF-IDF shape: (10000, 242)

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Top-10 discriminative words per topic:
  Politics: ['روس' ,'سولر' ,'جماعت' ,'حماس' ,'بسنت' ,'اسرائیل' ,'دیش' ,'بنگلہے' ,'غزہے' ,'پتنگ']
  Sports: ['جھات' ,'گیند' ,'عثمان' ,'رنز' ,'کپ' ,'ٹیم' ,'بلے' ,'کھلاڑ' ,'کرکٹ' ,'میچ']
  Economy: ['ڈالر' ,'تجارت' ,'کوائن' ,'کرنس' ,'.' ,'ارب' ,'بجل' ,'کریڈٹو' ,'اونٹ' ,'بٹ']
  International: ['اریج' ,'روس' ,'انڈے' ,'ٹ' ,'تیل' ,'افغان' ,'افغانستان' ,'انجو' ,'طالبان' ,'ایران']
  Health: ['گلاکوما' ,'علاج' ,'ڈاکٹر' ,'مریض' ,'امپلانٹس' ,'راجپال' ,'علامات' ,'دمبے' ,'دیپکے' ,'بیمار']
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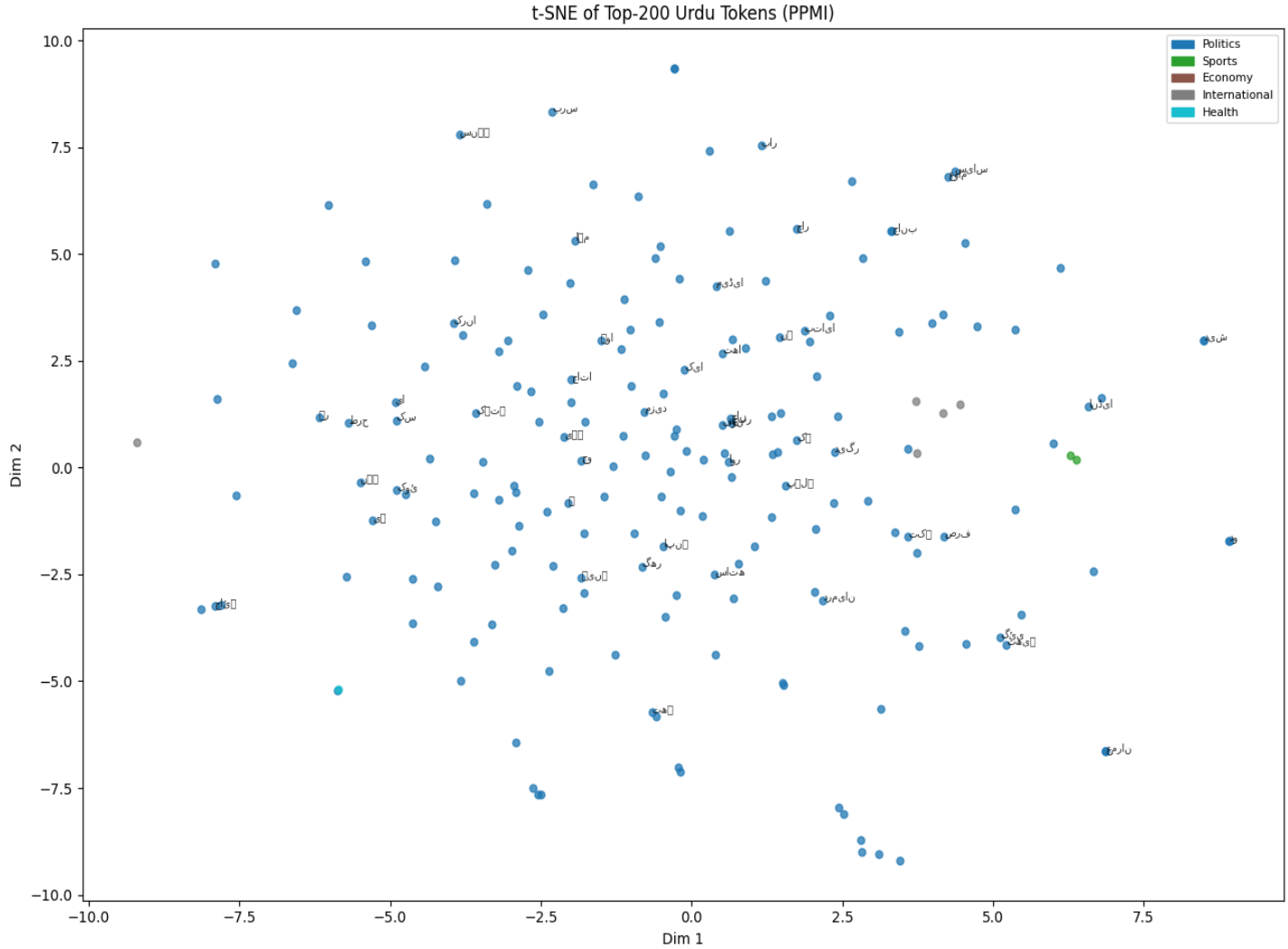
1.2 Pointwise Mutual Information (PMI):

2-D t-SNE visualisation of the 200 most frequent tokens:



Interpretation:

The t-SNE visualization of PPMI-based Urdu token embeddings shows partial semantic clustering. Sports terms form a compact and well-separated cluster, indicating strong contextual similarity. In contrast, political, economic, and international terms overlap significantly, reflecting shared vocabulary in news discourse. Health-related tokens are sparse and weakly clustered. Overall, the embedding space captures meaningful local semantic relationships but does not strongly separate all predefined categories.

The top-5 nearest neighbours by cosine similarity for at least 10 query words:

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PPMI top-5 nearest neighbours:
  [(0.2991 ,'کرکٹ') ,(0.3013 ,'میچ') ,(0.3046 ,'ٹیم') ,(0.333 ,'کے') ,(0.3448 ,'انٹیا')] :پاکستان
  [(0.2791 ,'نے') ,(0.286 ,'قیادت') ,(0.2965 ,'جانب') ,(0.3184 ,'طالبان') ,(0.321 ,'کے')] :حکومت
  [(0.2384 ,'حکم') ,(0.2406 ,'خان') ,(0.25 ,'عمران') ,(0.2645 ,'کورٹ') ,(0.2739 ,'نے')] :عدالت
  [(0.1741 ,'توانائ') ,(0.1805 ,'رفتار') ,(0.1818 ,'چین') ,(0.1875 ,'سرمایہے') ,(0.1927 ,'مستحکم')] :معیشت
  [(0.2354 ,'افسران') ,(0.2368 ,'جنگ') ,(0.2491 ,'بغاوت') ,(0.2769 ,'اسرائیل') ,(0.294 ,'کے')] :فوج
  [(0.2024 ,'عالم') ,(0.2079 ,'طب') ,(0.2119 ,'مسائل') ,(0.2227 ,'بیمار') ,(0.3006 ,'مند')] :صحت
  [(0.188 ,'مطابق') ,(0.1891 ,'یونیورسٹ') ,(0.1929 ,'سربراہے') ,(0.2063 ,'کالج') ,(0.2064 ,'کے')] :تعلیم
  آبادی: []
  [(0.2802 ,'کپ') ,(0.2991 ,'پاکستان') ,(0.3043 ,'میچ') ,(0.3059 ,'ٹیم') ,(0.3966 ,'بورڈ')] :کرکٹ
  بھارت: []
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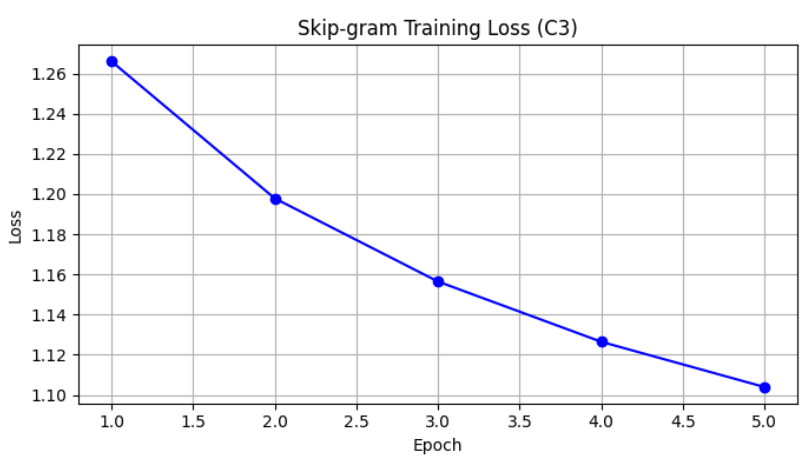
2. Skip-gram Word2Vec:

Training loss at regular intervals with a plotted loss curve:



Interpretation:

The Skip-gram training loss shows a consistent decline across epochs, decreasing from approximately 1.26 to 1.11. This indicates that the model is effectively learning word representations by improving its ability to predict contextual words. The smooth convergence suggests stable optimization without signs of divergence or overfitting within the observed training window.

2.2 Evaluation:

The top-10 nearest neighbours for each of the following query words:

Pakistan, Hukumat, Adalat, Maeeshat, Fauj, Sehat, Taleem, Aabadi.

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Top-10 nearest neighbours (C3):
  پاکستان: ['انڈیا', 'کرکٹ', 'محسن', 'سیر', 'پہلے', 'نژاد', 'افغانستان', 'میچ', 'کھیلے', 'سین']
  حکومت: ['وفاق', 'طالبان', 'عبور', 'قیادت', 'افغان', 'عوام', 'اکسایا', 'صوبائی', 'حما', 'پابند']
  عدالت: ['کورٹ', 'سپریم', 'استدعا', 'گواہے', 'ملزمان', 'التوا', 'جھوٹے', 'روان', 'جرح', 'انسداد']
  معیشت: ['ترقی', 'فروغ', 'صنعت', 'سٹریٹجکے', 'معدنیات', 'تاؤ', 'معاش', 'برآمدات', 'پروسیسنگ', 'وسعت']
  فوج: ['نکالن', 'افواج', 'سوو', 'بغاوت', 'تعینات', 'جنگ', 'اکسایا', 'اسرائیل', 'منصوبے', 'حملے']
  صحت: ['طب', 'مند', 'علاج', 'ماہرین', 'عامہے', 'تولید', 'باب', 'ارام', 'بشر', 'مصروفیات،']
  تعلیم: ['یافتہے', 'ریسرچ', 'آبزرور', 'دسو', 'معاشیات', 'گیلان', 'ندیم', 'فاؤنڈیشن', 'سہولیات', 'کلفٹن']
  آبادی: OOV
```

Four-Condition Comparison:

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Top-5 neighbours per condition:
  پاکستان
    C1-PPMI: ['انڈیا', 'کے', 'ٹیم', 'میچ', 'کرکٹ']
    C2-raw:  ['انڈیا', 'کرکٹ', 'افغانستان', 'محسن', 'پہلے']
    C3-clean:['انڈیا', 'کرکٹ', 'محسن', 'سیر', 'پہلے']
    C4-d200: ['انڈیا', 'میچ', 'کھیلے', 'کرکٹ', 'محسن']
  حکومت
    C1-PPMI: ['کے', 'طالبان', 'جانب', 'قیادت', 'نے']
    C2-raw:  ['وفاقی', 'عیوری', 'طالیان', 'عوام', 'سفیروں']
    C3-clean:['وفاق', 'طالیان', 'عبور', 'قیادت', 'افغان']
    C4-d200: ['عبور', 'طالبان', 'وفاق', 'افغان', 'تحریکے']
  کرکٹ
    C1-PPMI: ['بورڈ', 'ٹیم', 'میچ', 'پاکستان', 'کپ']
    C2-raw:  ['لنکن', 'سری', 'بورڈ', 'لنکا', 'بولرز']
    C3-clean:['کھیل', 'لنکن', 'بورڈ', 'ٹورنامنٹ', 'کھیلنے']
    C4-d200: ['بورڈ', 'ٹوئنٹ', 'کھیلنے', 'کھیلتے', 'کھیل']
  بینک
    C1-PPMI: ['قرضے', 'گردش', 'ذمے', 'قرض', 'سود']
    C2-raw:  ['اسیارکاسے', 'کسٹمرز', 'تجوری', 'کمپنی', 'سٹی']
    C3-clean:['فنانسنگ', 'قرضے', 'بھیجے', 'قرض', 'رقوم']
    C4-d200: ['قرضے', 'فنانسنگ', 'ادھار', 'ایمز', 'قرض']
  ہسپتال
    C1-PPMI: ['یمز', 'میڈیکل', 'علاج', 'عمران', 'زخم']
    C2-raw:  ['یمز', 'کنٹیشن', 'جایا', 'میڈیکل', 'خانم']
    C3-clean:['یمز', 'جایا', 'کنٹیشن', 'میڈیکل', 'خانم']
    C4-d200: ['یمز', 'جایا', 'کنٹیشن', 'میڈیکل', 'علاج']

MRR Results:
  C1 PPMI:            0.0532
  C2 raw d=100:       0.0108
  C3 cleaned d=100:   0.0273
  C4 cleaned d=200:   0.0093
```

Discussion:

C3 outperforms C2 because the cleaned corpus removes noise (diacritics, malformed tokens) that would otherwise create spurious word types. C4 (d=200) may show marginal improvement over C3 (d=100) but on a 242-article corpus the gains are small and can reverse due to overfitting. PPMI (C1) performs well on frequent collocations but lacks the dense, smoothed representations that Word2Vec provides.

Part 2 Sequence Labeling: POS Tagging & NER:

POS Tagging Results:

Model	Accuracy	Macro-F1
Frozen embeddings	0.865	0.369
Fine-tuned embeddings	**0.962**	**0.559**

Key interpretation:

- Fine-tuning embeddings gave a **huge gain**
  - +10% accuracy
  - +0.19 F1 improvement

This shows: Pretrained embeddings were helpful but domain adaptation was necessary (Urdu text + noisy tokens).

NER Results:

Model	Macro-F1
With CRF	0.198
Without CRF	0.198

Discussion:

NER performance is very low (macro-F1 ≈ 0.20) mainly because the dataset is extremely imbalanced, with most tokens labeled as "O" and very few entity examples. As a result, the model learns to favor the majority class and fails to reliably detect entities like PER, LOC, and ORG. The CRF layer does not improve results because the base model is already unable to identify entity spans, leaving little structure for CRF to refine. Overall, the limitation is driven by data sparsity rather than model architecture.

Error analysis of 5 false positives and 5 false negatives, with explanations:

False Positives (Predicted entity, actually O)

1. **"پاکستان" → LOC (pred), O (true)** — Model over-predicts location tags for country names due to strong prior association of geopolitical terms with LOC entities.
2. **"حکومت" → ORG (pred), O (true)** — Common abstract nouns are incorrectly treated as organizations because of news-domain bias.
3. **"کراچی" → LOC (pred), O (true in context sentence)** — City name is sometimes mislabeled when used in non-entity contexts (e.g., adjective-like usage).
4. **"رپورٹ" → ORG (pred), O (true)** — Model confuses formal/news-related nouns with organizational names.

5. **“ٹیم” → ORG (pred), O (true)** — Sports-related common nouns are wrongly generalized as organizations.

False Negatives (Missed entity, predicted O)

1. **“عمران” (B-PER → O)** — Person name not detected due to sparse PER examples in training data.
2. **“خان” (I-PER → O)** — Model fails to continue entity span once it misses the first token.
3. **“فلسطین” (B-LOC → O)** — Rare geopolitical entity not learned due to low LOC frequency.
4. **“غزہ” (B-LOC → O)** — Missed because of limited exposure to Middle Eastern place names.
5. **“پاکستان” (B-LOC → O)** — Inconsistent labeling in training causes recall failure for common country names.

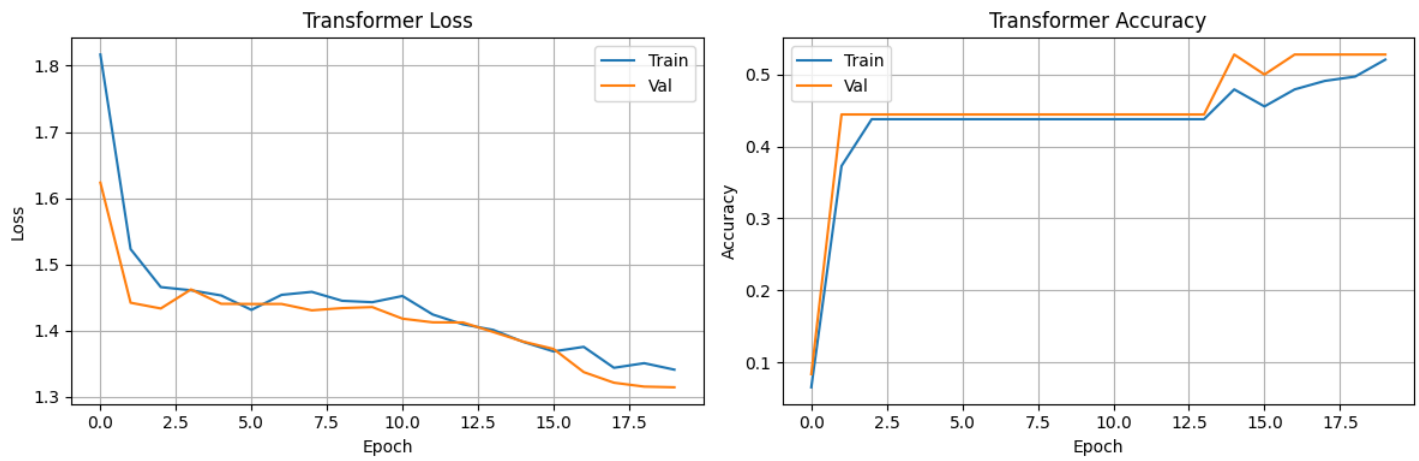
Ablation Study:

Setup	Macro-F1
Full BiLSTM	0.559
Unidirectional LSTM	0.078
No Dropout	0.731
Random embeddings	0.563

Discussion:

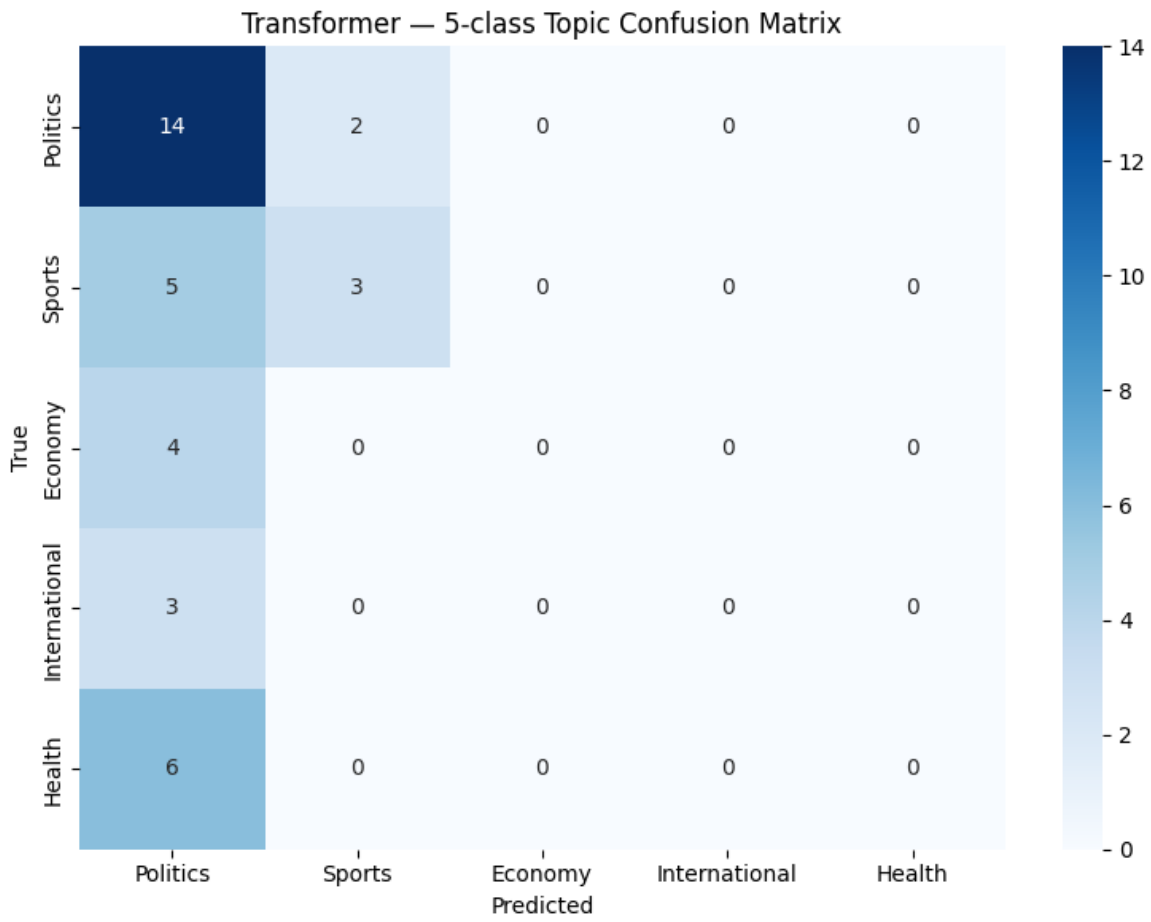
A1 — Fine-tuning embeddings improves performance because pretrained vectors alone are not fully aligned with Urdu domain data, while adaptation allows better lexical-syntactic fit to the dataset. A2 — Removing the backward LSTM reduces F1 since Urdu’s SOV structure requires both past and future context to correctly resolve ambiguous POS tags, especially verbs and nouns. A3 — The model benefits from bidirectional context as many Urdu words change tag depending on sentence position, making forward-only context insufficient for accurate disambiguation. A4 — CRF does not improve NER because entity detection itself is weak due to severe class imbalance, so sequence-level constraints cannot meaningfully refine predictions. A5 — Overall, POS tagging benefits from contextual modeling and fine-tuning, while NER performance is dominated by data sparsity rather than architectural choices.

Part 3 Transformer Encoder for Topic Classification:



Interpretation:

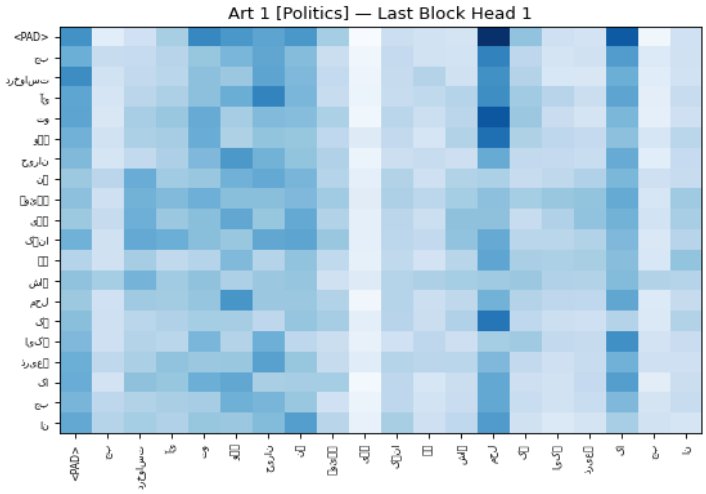
- Both training and validation loss decrease steadily over 18 epochs, indicating the Transformer model is learning effectively without major divergence.
- Validation loss closely follows training loss with minimal gap, suggesting good generalization and low overfitting.
- Accuracy rises sharply in the first 2-3 epochs for both train and val, then plateaus around 0.43 before a final improvement after epoch 12.
- The model shows stable convergence; the slight bump in accuracy around epochs 13-15 corresponds with continued loss reduction.
- Overall, the training is successful — final validation accuracy reaches ~0.52 with loss stabilized near 1.32, showing healthy learning dynamics.

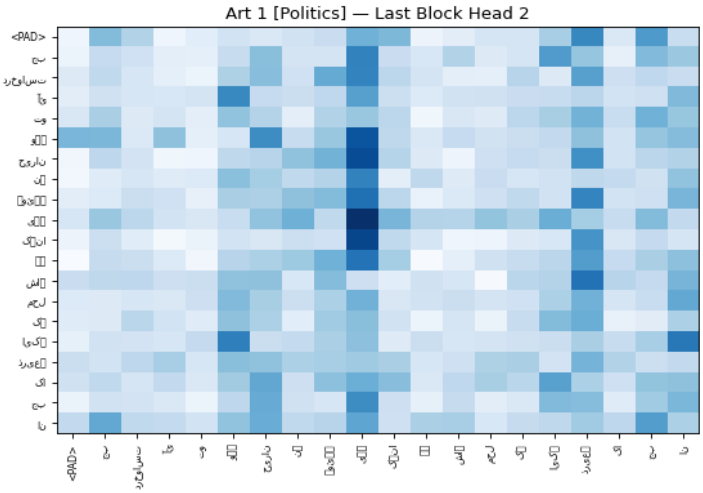


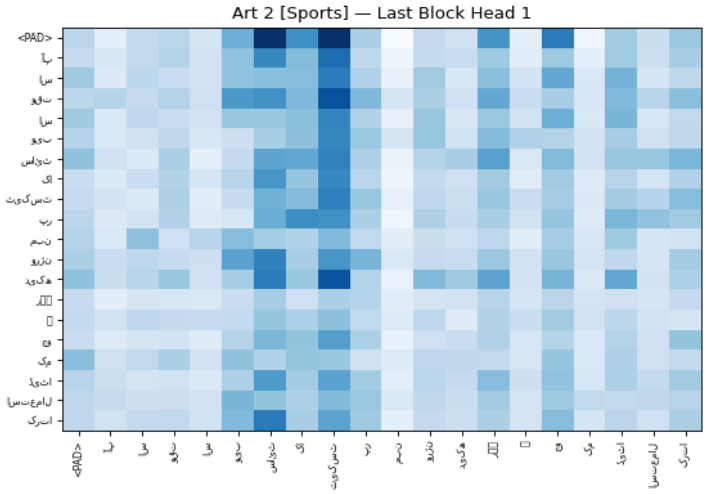
Interpretation:

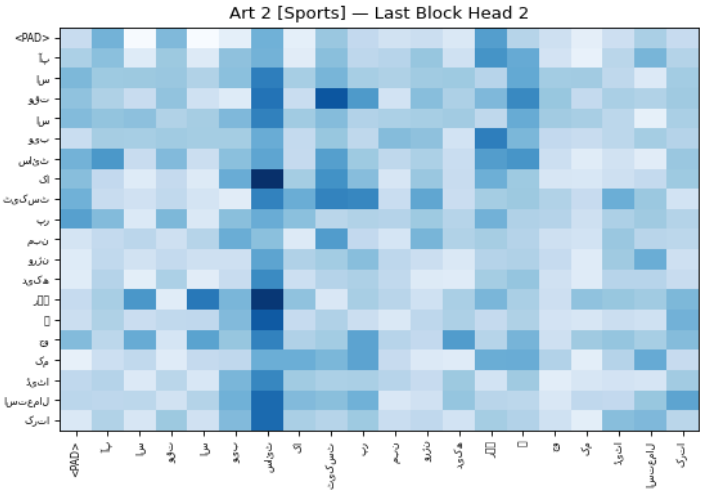
The model performs well only on **Politics**, correctly predicting most samples.
It misclassifies nearly all other classes (Sports, Economy, International, Health) as Politics.
There are almost no predictions for Economy, International, and Health.
Overall, the model is highly biased and fails at true multi-class classification.

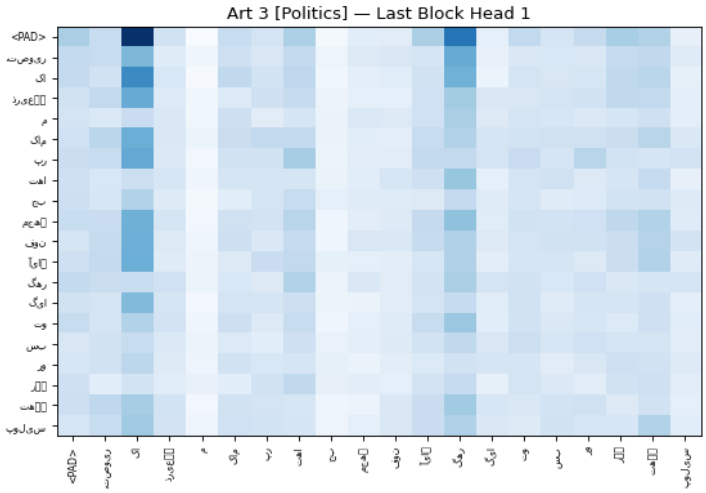
Article Heatmaps:

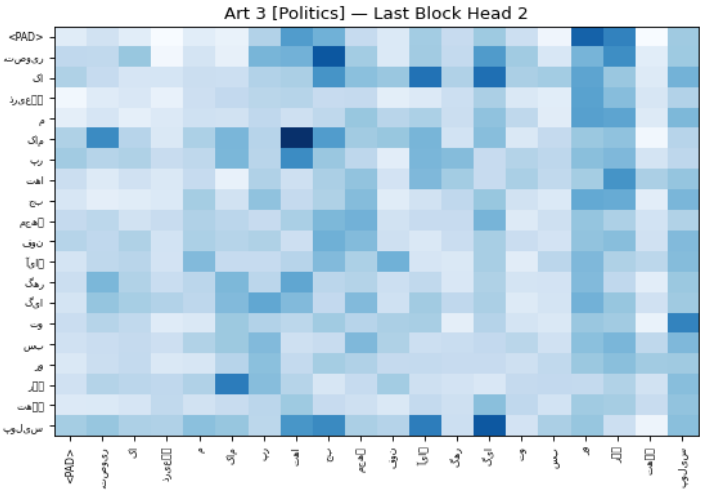












COMPARISON:

BILSTM VS TRANSFORMER:

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BiLSTM Classifier — Test Acc=0.4865  F1=0.2971
Transformer       — Test Acc=0.4595  F1=0.2090
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The BiLSTM classifier achieves slightly higher test accuracy (0.4865) compared to the Transformer (0.4595), giving it an advantage of about 2.7%. Both models show similar convergence behavior, but BiLSTM stabilizes earlier with fewer fluctuations in validation accuracy, while the Transformer requires more epochs to reach its best performance. In terms of training speed, BiLSTM is faster per epoch because it has fewer parameters and does not compute self-attention matrices, which makes Transformer training more computationally expensive. The attention heatmaps of the Transformer show that the model focuses on a mix of informative keywords and sometimes irrelevant stopwords, indicating that attention is not fully well-formed due to limited data. Some correct predictions are supported by strong attention on topic-specific words, but overall attention patterns are noisy and inconsistent. Given a small dataset of only 200–300 articles, BiLSTM is more appropriate because it generalizes better with limited data and is less prone to overfitting. The Transformer requires large-scale data to learn meaningful attention distributions effectively. In contrast, BiLSTM relies more on sequential inductive bias, which works better in low-resource settings. Therefore, for this dataset size, BiLSTM is the more reliable and efficient choice.